

ZeroAD: Zero-Shot Industrial Anomaly Detection via Multi-Scale CLIP Feature Scoring VAND 4.0 Challenge – Industrial Track (Zero-Shot)

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Abstract

We present ZeroAD, a parameter-efficient zero-shot anomaly detection method for the VAND 4.0 Industrial Track (Zero-Shot category). The core idea is to leverage the rich visual representations learned by a frozen CLIP ViT-L/14@336px backbone and train only a minimal set of linear scoring heads for anomaly detection. Specifically, we extract multi-scale features from 5 intermediate CLIP transformer layers and attach 10 independent linear projections ($768 \rightarrow 1$) — 5 for image-level classification and 5 for pixel-level segmentation — resulting in only 7,690 trainable parameters. Training is performed on the test split of MVTec AD using cross-entropy and Focal/Dice losses. **No images from the MVTec AD 2 dataset are used at any stage — neither for training, validation, nor hyperparameter tuning.** On MVTec AD 2 test_public, our method achieves a mean SegF1 of 22.9%. The code-base, model weights, and all training configurations are publicly available at <https://github.com/clearable/ZeroAD-VAND2026>.

1. Participant Information

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2. Submission Information

- **Track:** Industrial
- **Industrial Track?** Yes
- **Zero-Shot?** Yes
- **Retail Track?** No
- **Off-the-shelf VLM submission?** No

- **Code Repository:** <https://github.com/clearable/ZeroAD-VAND2026>

3. Introduction

3.1. Background

Anomaly detection in industrial manufacturing is essential for automated quality inspection. Detecting surface defects, structural anomalies, and contamination in products requires both image-level classification (“is this product defective?”) and pixel-level localization (“where is the defect?”). Traditional supervised approaches demand large, category-specific labeled datasets, which are expensive to collect and do not generalize across product types.

Vision-language models such as CLIP [4] have recently emerged as powerful feature extractors that learn generalizable visual representations from hundreds of millions of image-text pairs. Several works [5] have demonstrated that CLIP features can be adapted for anomaly detection. However, many existing approaches either require complex prompt engineering, large learnable modules, or access to target-domain data, limiting their applicability in true zero-shot settings.

Our work is motivated by a key observation: **CLIP’s intermediate-layer features already encode sufficient information to distinguish normal from anomalous regions.** Rather than designing complex adapters or decoders, we hypothesize that a single linear projection per feature layer suffices to extract anomaly signals. This leads to an extremely parameter-efficient design that is inherently resistant to overfitting and generalizes well across domains.

3.2. Challenge Description

The VAND 4.0 Industrial Track evaluates anomaly segmentation methods on the MVTec AD 2 dataset [1], which introduces 8 new industrial product categories (can, fabric, fruit_jelly, rice, sheet_metal, vial, wallplugs, walnuts) with

multiple test splits. The primary evaluation metric is pixel-level SegF1 (F1 score at the optimal threshold).

The **Zero-Shot category** imposes the following constraint: *no images from the MVTec AD 2 dataset may be used during training*. This tests whether the method can generalize to entirely unseen product types and defect patterns without any adaptation.

4. Methodology

4.1. Model Design

Approach. Our approach is built on three design principles:

1. **Freeze everything possible:** The CLIP backbone contains 304M parameters trained on web-scale data. Fine-tuning it on a small anomaly detection dataset would destroy these generalizable representations and cause severe overfitting.
2. **Minimize learnable parameters:** By using only linear layers (768→1), we reduce the risk of memorizing source-domain-specific patterns that do not transfer to the target domain.
3. **Exploit multi-scale features:** Different transformer layers capture different levels of abstraction. Shallow layers (block 12) encode texture and edge information; deep layers (block 24) encode semantic content. Using 5 layers provides complementary anomaly signals.

We also conducted extensive ablation studies to validate these principles (see Section 7.5).

Architecture. The ZeroAD model consists of three components:

1. **Frozen CLIP ViT-L/14@336px Encoder.** We use the visual encoder from CLIP ViT-L/14@336px (OpenAI release). The image is resized to 518×518, yielding a 37×37 patch grid (1,369 patches). We extract features from 5 intermediate transformer blocks: layers 12, 15, 18, 21, and 24. Each layer’s output passes through the frozen `ln_post` layer normalization and `proj` linear projection (1024→768), producing:
 - CLS token: $\mathbf{c}^{(i)} \in \mathbb{R}^{768}$
 - Patch tokens: $\mathbf{P}^{(i)} \in \mathbb{R}^{1369 \times 768}$
 Both are L2-normalized before scoring: $\hat{\mathbf{c}}^{(i)} = \mathbf{c}^{(i)} / \|\mathbf{c}^{(i)}\|_2$.
2. **Image-Level Anomaly Scorers (5 layers).** For each layer $i \in \{1, \dots, 5\}$, a linear layer $W_{\text{cls}}^{(i)} \in \mathbb{R}^{768 \times 1}$ maps the normalized CLS token to a scalar anomaly logit:

$$s_{\text{cls}}^{(i)} = W_{\text{cls}}^{(i)\top} \hat{\mathbf{c}}^{(i)} + b_{\text{cls}}^{(i)} \quad (1)$$

The output is paired with a fixed zero (representing “normal”) to form a 2-class logit: $\mathbf{l}_{\text{cls}}^{(i)} = [0, s_{\text{cls}}^{(i)}]$.

3. **Pixel-Level Anomaly Scorers (5 layers).** Similarly, for each layer i , a linear layer $W_{\text{seg}}^{(i)} \in \mathbb{R}^{768 \times 1}$ scores each patch independently:

$$\mathbf{S}_{\text{seg}}^{(i)} = \text{Reshape}_{37 \times 37} \left(W_{\text{seg}}^{(i)\top} \hat{\mathbf{P}}^{(i)} + b_{\text{seg}}^{(i)} \right) \quad (2)$$

The result is upsampled to 518 × 518 via bilinear interpolation.

Total trainable parameters: $5 \times 2 \times (768 + 1) = 7,690$ — less than 0.008M.

Prompting / Training. This method does **not** use text prompts or VLM-based prompting. It is a pure vision approach.

Training data: The test split of MVTec AD [2] (15 categories), which contains both normal and anomalous images with pixel-level ground truth masks. **No MVTec AD 2 data is used.**

Loss function: The total loss combines image-level cross-entropy and pixel-level Focal/Dice losses across all 5 layers:

$$\mathcal{L} = \underbrace{\sum_{i=1}^5 \text{CE}(\mathbf{l}_{\text{cls}}^{(i)}, y)}_{\text{image-level}} + \lambda \underbrace{\sum_{i=1}^5 [\mathcal{L}_{\text{focal}}^{(i)} + \mathcal{L}_{\text{dice,anom}}^{(i)} + \mathcal{L}_{\text{dice,norm}}^{(i)}]}_{\text{pixel-level}} \quad (3)$$

where y is the image-level binary label, $\lambda = 4$ (following AnomalyCLIP [5]), and the pixel-level losses operate on softmax-normalized logit pairs $[0, s]$. Focal loss [3] addresses class imbalance (anomalous pixels are rare), while Dice loss encourages spatial overlap with ground truth masks.

Training configuration:

- Optimizer: Adam (lr = 10^{-3} , $\beta = (0.5, 0.999)$)
- Batch size: 8
- Epochs: 15
- Data augmentation: None
- Learning rate scheduler: None (constant)
- Random seed: 111
- Image size: 518 × 518 (bicubic resize + center crop)
- Normalization: CLIP standard (mean = [0.481, 0.458, 0.408], std = [0.269, 0.261, 0.276])

4.2. Inference Pipeline

1. Extract L2-normalized CLS and patch features from 5 CLIP layers.
2. Compute per-layer image-level logit $s_{\text{cls}}^{(i)}$ and pixel-level map $\mathbf{S}_{\text{seg}}^{(i)}$.
3. Average across 5 layers: $\bar{s}_{\text{cls}} = \frac{1}{5} \sum_i s_{\text{cls}}^{(i)}$ and $\bar{\mathbf{S}}_{\text{seg}} = \frac{1}{5} \sum_i \mathbf{S}_{\text{seg}}^{(i)}$.

4. Compute anomaly map via the formula from Anomaly-CLIP [5]:

$$\mathbf{M} = \frac{\bar{\mathbf{S}}_{\text{seg}}^{(\text{anom})} + 1 - \bar{\mathbf{S}}_{\text{seg}}^{(\text{norm})}}{2} \quad (4)$$

5. Fuse image-level and pixel-level scores:

$$s_{\text{final}} = (1 - \lambda_p) \bar{s}_{\text{cls}} + \lambda_p \max(\mathbf{M}), \quad \lambda_p = 0.5 \quad (5)$$

6. Apply Gaussian smoothing ($\sigma = 4$) to the anomaly map $\mathbf{M}$.

5. Datasets and Evaluation

5.1. Benchmark Dataset Utilization

Training dataset: MVTec AD [2] test split — 15 industrial product categories with pixel-level defect annotations. This dataset is publicly available and is **entirely separate from MVTec AD 2**.

Preprocessing:

- Resize to 518×518 using bicubic interpolation
 - Center crop to 518×518
 - Normalize with CLIP statistics
 - Ground truth masks binarized at threshold 0.5
- No data augmentation** is applied. No validation set is used; all hyperparameters are fixed a priori.

5.2. Evaluation Metrics

We evaluate on MVTec AD 2 `test_public` (8 categories) using:

- **SegF1** (primary, per competition): pixel-level F1 at optimal threshold
- I-AUROC, I-AP: image-level classification metrics
- P-AUROC, P-AUPRO: pixel-level localization metrics

5.3. Evaluation Setup — Zero-Shot Compliance

Our method strictly adheres to the zero-shot protocol defined by the VAND 4.0 Industrial Track:

1. **No MVTec AD 2 data used at any stage.** Not for training, not for validation, not for hyperparameter tuning, not for threshold selection.
2. **The model is not updated during inference.** It remains in its initial trained state. No test-time adaptation or on-line learning is performed.
3. **No information from target-category test samples is used during inference** for any other test sample. Each image is processed independently.
4. **Pre-training strategy:**
 - Backbone: CLIP ViT-L/14@336px from OpenAI
 - Pretrained on: LAION-400M / OpenAI WebImage-Text (web-scraped image-text pairs)
 - No modifications to the pretrained weights
5. **Training data:** MVTec AD test split only (15 categories, publicly available)

6. **All hyperparameters are fixed** (lr, epochs, batch size, λ , σ) and were not tuned on MVTec AD 2.

6. Results

6.1. Performance Metrics

Table 1 shows per-category SegF1 on MVTec AD 2 `test_public` for both training sources.

Table 1. SegF1 (%) on MVTec AD 2 `test_public`. Two training sources compared.

Category	MVTec AD trained	VisA trained
can	0.4	0.1
fabric	7.1	1.5
fruit_jelly	32.4	8.8
rice	43.8	0.6
sheet_metal	15.6	1.5
vial	29.7	13.9
wallplugs	8.0	3.7
walnuts	46.0	6.5
Mean	22.9	4.6

6.2. Comparison and Analysis

The MVTec AD-trained model outperforms the VisA-trained variant by a large margin (22.9% vs. 4.6% mean SegF1). This is attributable to the domain similarity between MVTec AD and MVTec AD 2: both focus on industrial surface defects with similar imaging conditions.

Per-category observations:

- **Best:** walnuts (46.0%) and rice (43.8%) — textured surfaces where anomalies manifest as texture irregularities that CLIP features capture well.
- **Moderate:** fruit_jelly (32.4%), vial (29.7%) — mixed defect types.
- **Challenging:** can (0.4%) — smooth metallic surfaces where defects are subtle color/reflectance changes, difficult for frozen CLIP features to detect without task-specific adaptation.

7. Discussion

7.1. Challenges and Solutions

Challenge 1: Domain gap. The training domain (MVTec AD) and target domain (MVTec AD 2) contain entirely different product categories. We address this by:

- Using frozen CLIP features that are domain-agnostic by design (pretrained on 400M+ diverse images).
- Training only 7,690 parameters to minimize the risk of memorizing source-domain-specific patterns.
- Using multi-scale features (5 layers) for robustness.

Challenge 2: Softmax suppression. In early ablation experiments (#9 vs #11 in our logs), we found that ap-

plying softmax to the 2-class logits $[0, s]$ degrades pixel-level performance. While softmax does not affect ranking metrics (AUROC), it compresses the logit range and hurts threshold-dependent metrics like SegF1. **Solution:** Remove softmax and use raw logits directly.

Challenge 3: Scorer complexity. We tested an MLP scorer (768→512→1, #8) and found it performed comparably on image-level metrics but worse on pixel-level P-AUPRO (62.4% vs 87.5%). The additional capacity enables overfitting to source-domain pixel patterns. **Solution:** Use the simplest possible scorer (single linear layer).

7.2. Model Robustness and Adaptability

The model exhibits several desirable robustness properties:

- **Frozen backbone** prevents catastrophic forgetting of CLIP’s pretrained knowledge.
- **Extreme parameter efficiency** (7,690 params) provides strong regularization.
- **Multi-layer fusion** averages over 5 independent scoring heads, providing ensemble-like robustness.
- **No data augmentation or scheduling** reduces the number of “hidden” hyperparameters.

The method adapts to new domains by simply retraining the 10 linear layers on a different source dataset (e.g., MVTec AD vs. VisA), which takes approximately 5 minutes on a single GPU.

7.3. Model Efficiency

Table 2. Computational cost of ZeroAD.

Metric	Value
Trainable parameters	7,690 (0.008M)
Total parameters (incl. frozen CLIP)	304M
Training time (15 epochs, single GPU)	~5 min
Inference speed (RTX 3090)	~15 img/s
Checkpoint size	38 KB
CLIP model size (auto-downloaded)	890 MB
Peak GPU memory (inference, bs=1)	~4 GB

7.4. Zero-Shot Compliance

As required by the VAND 4.0 Industrial Track Zero-Shot rules:

1. **Pre-training strategy:** We use the publicly released CLIP ViT-L/14@336px model from OpenAI. The model was pretrained on a large-scale web-scraped image-text dataset (LAION-400M / WebImageText). We do not modify the pretrained weights in any way.
2. **Confirmation: No MVTec AD 2 data was used during training.** Our training pipeline loads only the MVTec AD dataset (15 categories). The MVTec AD 2 dataset (8 categories) is used exclusively for evaluation after training is complete. No images, labels, or statistics from MVTec AD 2 influence the model in any way.

3. **Model specification:** CLIP ViT-L/14@336px, OpenAI release. Downloaded from <https://openaipublic.azureedge.net/clip/models/>. SHA256 checksum: 3035c92b.... No fine-tuning of backbone weights.

7.5. Ablation Studies

During development, we conducted systematic ablation studies on the zero-shot framework. Key findings (evaluated on MVTec AD cross-domain):

Table 3. Key ablation results (VisA→MVTec, zero-shot).

Variant	I-AUC	I-AP	P-AUC	P-PRO
MLP 768→512→1 (#8)	89.1	93.8	88.6	62.4
Linear 768→1 + softmax (#9)	88.6	93.9	91.2	84.7
Dual scorers + softmax (#10)	89.0	94.0	91.3	72.2
Linear 768→1, no softmax (#11)	89.5	94.3	91.1	87.5

Key takeaways:

- Removing softmax improved P-AUPRO from 84.7% to 87.5% (#9→#11).
- Simpler scorer (linear) outperformed MLP on P-AUPRO by +25.1% (#8 vs #11).
- Using two independent scorers for normal/anomaly (#10) hurt P-AUPRO vs. fixing normal logit to 0 (#11).
- **Simplicity wins:** the least complex variant (#11) achieved the best overall results.

7.6. Future Work

- **Learnable layer fusion:** Replace simple averaging with learned attention over 5 layers.
- **Category-adaptive thresholding:** Learn per-category thresholds on a validation set to optimize SegF1.
- **Few-shot extension:** When a small number of reference images are available, combine with PatchCore-style nearest-neighbor matching (our #12 experiment achieved 91.2% I-AUROC with 8-shot).
- **Larger backbones:** Explore ViT-G or SigLIP for richer features.

8. Conclusion

ZeroAD demonstrates that effective zero-shot anomaly detection can be achieved with extreme parameter efficiency. By freezing a CLIP ViT-L/14@336px backbone and training only 7,690 parameters (10 linear layers), we achieve 22.9% mean SegF1 on MVTec AD 2 `test_public` without using any target-domain data. Through systematic ablation studies, we showed that simplicity is key: a single linear projection per layer outperforms MLP variants, dual-scorer designs, and softmax normalization. The lightweight design (38 KB checkpoint, 5-minute training) enables rapid deployment to new inspection scenarios.

Zero-shot declaration: No images from the MVTec AD 2 dataset were used during training. The model relies solely on frozen CLIP pretrained features and 10 lightweight linear scoring heads trained on MVTec AD.

References

- [1] Karl Batzner, Lars Heckler, and Rebecca König. Beyond MVTec AD: A multi-setting benchmark for anomaly detection in the wild. *arXiv preprint arXiv:2503.21622*, 2025. [1](#)
- [2] Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger. Mvtec ad—a comprehensive real-world dataset for unsupervised anomaly detection. *International Journal of Computer Vision*, 129:1038–1059, 2021. [2](#), [3](#)
- [3] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)*, 2017. [2](#)
- [4] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021. [1](#)
- [5] Qihang Zhou, Guansong Pang, Yu Tian, Shibo He, and Jiming Chen. Anomalyclip: Object-agnostic prompt learning for zero-shot anomaly detection. In *International Conference on Learning Representations (ICLR)*, 2024. [1](#), [2](#), [3](#)